

Ashmit Mukherjee

New York University Abu Dhabi

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Education

New York University Abu Dhabi

Expected May 2027

B.S. in Computer Science. Studied at NYU Abu Dhabi, NYU Paris, and NYU New York.

Relevant coursework: Natural Language Processing; Principles of Data Science; Statistics for Social & Behavioral Sciences; Data Structures & Algorithms; Computer Systems Architecture; Operating Systems.

Research Interests

I build and evaluate AI systems using data, statistical inference, machine learning, experimental design, and research engineering. My work spans human-AI collaboration, parameter-efficient model adaptation, and multilingual NLP.

Publications and Working Papers

Benjamin Rosche, **Ashmit Mukherjee**, and Hanan Salam. “AI and Human Collaboration in Social Dilemmas: Strategic Interpretation, Normative Interpretation, and Collective Action in Shared Supply Chains.” *Working paper*, June 2026.

Ashmit Mukherjee and collaborators. “Bio-Informed LoRA for Signal Peptide Prediction.” Co-first author. *Under review at an EMNLP 2026 workshop*, 2026.

Research Experience

Research Assistant: Human-AI Collaboration, NYU Abu Dhabi

Feb 2026–Present

- Co-developing an online experiment on how AI-mediated interpretation and coordination affect collective action in shared global supply chains, with Benjamin Rosche and Hanan Salam.
- Designed a repeated sourcing game in which five human buyers choose fair pay, auditing, information-sharing links, communication, and threshold-dependent remediation around one supplier governed by a fixed decision rule.
- Developed a group-level, four-arm treatment design comparing structured information with strategic interpretation AI, normative interpretation AI, and collective action AI.
- Specified behavioral outcomes and mechanism measures for cooperation, responsibility attribution, behavioral homogenization, coalition formation, and persistence following AI withdrawal.

Research Assistant: eBRAIN Lab, NYU Abu Dhabi

Jan 2026–Present

- Co-developed **Bio-Informed LoRA**, a parameter-efficient fine-tuning method that injects residue-level biological priors (BLOSUM62, hydrophobicity, Grantham) into LoRA adapters for ESM-2 protein language models on signal peptide prediction.
- Showed BLOSUM62-guided LoRA matches or exceeds full fine-tuning on SignalP6 benchmarks across MCC, cleavage-site exactness, and residue-level F1, while training approximately 3.6% of model parameters and using approximately 43% less peak GPU memory at the 3B-parameter scale.
- Designed multi-seed evaluation protocols and a one-factor sensitivity screen showing that several single-seed improvements did not replicate.
- Manuscript under review (see Publications and Working Papers).

Research Projects

Hinglish Named Entity Recognition Benchmark

Fall 2024

- Fine-tuned multilingual transformer models, mBERT and XLM-RoBERTa, on the COMI-LINGUA dataset for Hindi-English code-mixed NER.
- Compared against zero-shot GPT-4o and Claude-3.5-Sonnet baselines under a common evaluation protocol; fine-tuned XLM-RoBERTa achieved 78% entity-level F1 versus 76% for GPT-4o.
- Analyzed trade-offs between model scale, task-specific fine-tuning, generalization, and computational cost.

Data Science Salary Prediction Platform

Fall 2025

- Built a benchmarking pipeline for more than 15 regression models using over 3,000 job postings; tracked experiments with MLflow and compared MAE, RMSE, and R^2 .
- Used SHAP to interpret associations between model predictions and features including geography, firm size, and skill requirements.

Community-Level Crime Modeling

Spring 2025

- Modeled statistical associations between socioeconomic indicators and violent crime rates across 1,994 communities; the analysis does not make causal claims.
- Cross-validation yielded $R^2 = 0.85$ for the selected predictive model; presented results through an interactive Streamlit interface.

CAMP: Campus Asset Management Platform

Fall 2025

- Contributed to a team-built platform for campus resource allocation and scheduling under shared constraints, including role-based access and conflict resolution.
- Built RESTful APIs, integration testing, containerized deployment, and automated CI/CD workflows.

Industry Experience

Machine Learning Engineer Intern, Zeek

Jun–Aug 2025

- Developed and evaluated segmentation models on noisy real-world data, iterating on preprocessing and model selection for production reliability.
- Applied model interpretability techniques to study robustness and bias sensitivity, helping the team decide which model variants to deploy.
- Contributed to the team's evaluation workflow by adding standardized metrics and validation procedures to the model development cycle.

Technical Skills

Languages: Python, C++, C#, JavaScript

ML & NLP: PyTorch, Transformers (Hugging Face), Scikit-Learn, XGBoost, PyCaret, SHAP

Computational Modeling: NumPy, NetworkX, game-theoretic simulation

Data Analysis: Pandas, SciPy, Matplotlib, Seaborn, Plotly

Experimental Methods: Experimental design, regression, cross-validation, multi-seed evaluation, sensitivity analysis

Research Engineering: MLflow, Git, Docker, GitHub Actions, REST APIs, automated testing, MongoDB

Applications: Streamlit, React, Node.js, L^AT_EX